YUANJUN (JOHNNY) DAI

216-269-2394 • yxd429@case.edu • Cleveland, OH
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SUMMARY

AI Infrastructure and Applied AI Engineer, **PhD candidate** at CWRU. Specializes in the security, optimization, and observability of ML/LLM training and inference infrastructure. 8 research papers (7 first-author) at IEEE, ACM, and Springer; **2+ years of systems software engineering** at Cisco and Broadcom; 2+ years of **cross-functional** product ownership at Midea and Xiaomi.

EDUCATION

Ph.D. Candidate in Computer Science (AI Infrastructure) — 08/2019 – Sep. 2026 (exp.)
Case Western Reserve University

M.S. in Electrical and Computer Engineering (Computer Architecture)
University of Pittsburgh

B.S. in Electrical Engineering
Northeastern University

SKILLS

Languages	Python, C/C++
ML Frameworks	PyTorch (DDP/FSDP), vLLM, Megatron-LM, NCCL, BytePS, scikit-learn
Parallelism Paradigms	Data Parallel (DDP/FSDP), Tensor Parallel (TP), Pipeline Parallel (PP), Hybrid Parallel
Infrastructure	Kubernetes, Docker, CI/CD Pipelines (GitHub Actions), ArgoCD, Slurm
Systems & APIs	FastAPI, REST API, gRPC
Observability	eBPF (Kernel Tracing), Prometheus, Grafana, DCGM Exporter, Intel RAPL
Cloud & HPC	Lambda Labs HPC, Ohio Supercomputer Center, NSF FABRIC, CloudLab
Networking	SDN (Software-Defined Networking), P4, NPU/ASIC (Broadcom BCM SDK)

RESEARCH & SYSTEMS ENGINEERING EXPERIENCE

Research Assistant, AI Infrastructure & ML Systems — 08/2019 – Sep. 2026 (exp.)

Case Western Reserve University

Research focus: **security, optimization, and observability** of **ML/LLM** training and inference infrastructure.

DYNAMIX – Reinforcement Learning-Based Adaptive Batch Scheduling for Distributed ML | Paper / Code

- Designed a **PPO**-based **Reinforcement Learning (RL)** scheduler monitoring hardware signals (CPU/memory, bandwidth, retransmission rate via **eBPF** probes) and statistical signals (loss convergence, generalization) to dynamically adapt batch size
- Co-scales learning rate via **Linear Scaling Rule**; evaluated on **PyTorch DDP** and **Parameter Server (BytePS)**
- Achieves **46% wall-clock speedup** and **~6% accuracy gain** across **8–64 A100 GPUs** over fixed batch size baseline

PRISM – Priority-Based Selective Reliability for Distributed ML Training | Paper

- Designed a transport-layer library (**NCCL/Gloo → UDP**) hooking into **PyTorch DDP**'s AllReduce at bucket-ready events; classifies gradients by magnitude into high-priority (full **SACK**-based retransmission) and low-priority (local compensation via cached values)
- Supports kernel-bypass (**LibVMA**) and kernel-based (**GSO/GRO**) variants; achieves **8–15% throughput gain for CNNs and 15–21% for LLMs** over TCP, matching NDP latency at 144-worker scale

Scalable LLM Serving Infrastructure on Kubernetes | Code

- Independently designed and built end-to-end a **K8s** microservices platform (Gateway / Router / Worker) with **vLLM** inference
- **ArgoCD** GitOps, **Prometheus/Grafana** observability, and **CI/CD** pipelines across **50+ deployments**

Domain-Adapted LLMs for Clinical Risk Adjustment Coding | Paper

- Built a 115-label HCC multi-label classifier on highly imbalanced EHR data (MIMIC-V), improving long-tail performance via **inverse-frequency weighted binary cross-entropy loss** and **per-label threshold calibration**, achieving **0.775 macro-F1 / 0.742 micro-F1 / 0.9362 macro-AUC**.
- Trained **PubMedBERT** using **PyTorch DDP** on 2×NVIDIA RTX A6000 GPUs for efficient scaling, while fine-tuning **Qwen-7B** using distribution-aware **few-shot prompting** (5-shot with negative, frequent, and rare-label demonstrations) to mitigate hallucination and enforce structured HCC outputs with improved sensitivity to sparse labels; scaled full-parameter training via **NeMo + Megatron-LM** (TP=1, PP=1) with **PyTorch FSDP (ZeRO-3)**, **FlashAttention**, and **activation checkpointing**, enabling training under GPU memory constraints (avoiding OOM).

Priority-Based eBPF Network Flow Telemetry for AI / DML Infrastructure | Paper / Code

- Designed PSketch, the first in-kernel priority-aware sketch using **eBPF**; high-priority flows tracked via in-kernel hashmap (O(1)), low-priority via **3-layer Count-Min Sketch**; enables **O(n) Top-K detection** with sub-linear memory, voting-based collision resolution and atomic operations for thread-safe updates
- Achieves **96% Top-K accuracy, 96.4% retransmission recall**, and <1.1% throughput overhead at 10 Gbps on FABRIC testbed

High-Resolution eBPF System Resource Profiling for DML Workloads | Paper / Code

- Designed eHashPipe, a dual-pipeline **eBPF** framework with **O(n) Top-K tracking**: (1) HashPipe-based memory profiler with cascading slot eviction and ptr2size deallocation tracking, (2) per-thread CPU tracker via sched_switch tracepoint
- Delivers **10–14× finer temporal resolution** than top/htop, ~11ms latency, and >90% Top-K fidelity on commodity Linux servers

Training-Time Side-Channel Attacks on Distributed ML Systems | Paper / Code

- **Training-phase black-box attack** on **MXNet BSP** distributed training, reconstructing full DNN architectures via two unprivileged side channels: **Intel RAPL** (PP0+DRAM, 50μs via powercap/sysfs) and 0-payload network packet sequences (**Seq_T**) for tensor size extraction
- **Lookahead** (sliding-window SMA) + **Lookbehind** algorithms isolate forward passes; **band-pass filtering** segments individual layers; **Decision Tree** (entropy, 97.7%, 10-fold CV on 84K+ segments) and **BiLSTM** classify layer types; weighted-average **MLP ensemble** (weights ∝ 1/validation error) recovers Conv/FC dimensions
- Achieves **>97% layer-type accuracy**; validated across 20 DNN models (LeNet through VGG/ResNet families)

Scotch – Elastic SDN Overlay for Control-Plane Scalability under DDoS | Paper / Code

- Designed an **OvS**-based elastic overlay: when physical switch OFA is saturated, tunnels new flows to a **vSwitch** pool via **event-driven** Packet-In Edge messages; built a centralized **Ryu SDN** controller for **flow monitoring** and **load balancing** across vSwitches
- Separates small DDoS flows (terminated at vSwitches) from large flows (migrated to physical paths); achieves **>70% controller congestion reduction** on 34-node GENI Clos topology

INDUSTRY EXPERIENCE

Software Engineer II 06/2015 – 08/2016
Cisco Systems, Inc.

- Developed NPU/ASIC features for a 100 Gbps core router (Azure, AWS, Alibaba Cloud) using Broadcom BCM SDK.
- Contributed to QoS and ECMP modules, reducing forwarding latency under heavy load.

Software Engineer Intern 10/2014 – 05/2015
Broadcom (Emulex)

- Re-architected RTOS data acquisition with DMA + circular-buffer, **reducing CPU load by 30%**; ported BladeEngine ASIC management CLI from Linux to Windows Server.

Product Manager 09/2016 – 06/2018
Midea Group

- Led technical PM for **Fun Oven II** across ML model training, Docker deployment on Alibaba Cloud, mobile app, and culinary algorithm design; won **Red Dot Design Award**, debuted at 2018 AWE Show · Red Dot ↗ · PR Newswire ↗

Technical Product Manager 07/2018 – 01/2019
Xiaomi

- Defined IoT appliance product requirements via competitive analysis and user research; assessed technical feasibility and delivered architecture proposals to align engineering and business stakeholders.

PUBLICATIONS

8 papers total, 7 first-author, 5 published at IEEE CCNC, IEEE ICNC, ACM TOIT, and Springer SecureComm; 3 under review (incl. ACM TOPS). Full list: johnny-dai-git.github.io/portfolio

PATENTS

Oven Temperature Control Method, Device and Computer-Readable Storage Medium — Granted CN 108594898 B | Issued July 23, 2021 | Assignee: Midea Group Co., Ltd. | Dai Yuanjun et al.